SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
Student Name:	NAVEEN.S	**Reg. No.:**	192324248
Section / Batch:		**Date:**	

Code of Conduct

I NAVEEN.S / 192324248 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: _____

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Q4. During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log 10$ of the ratio).

Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Q28. A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

Q29. A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

Q30. An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

Q32. Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

Q33. A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

Q34. A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

Q35. A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

Q36. A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

Q37. A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

Q38. A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

Q39. A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

Q40. A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

Q41. A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

Q42. An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

Q43. A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

Q44. A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

Q45. A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

Q46. A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

Q47. A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

Q48. A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

Q49. A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

Q50. A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = πR^2) and needs to blanket a 350 km² service region. Roughly how many towers will the rollout require?

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 1

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

SOLUTION

Formula: Full-duplex channels = Total bandwidth / bandwidth per simplex channel / 2.

Substitution: 45 MHz = 45,000 kHz. Simplex channels = 45,000 / 25 = 1,800. Full-duplex channels = 1,800 / 2 = 900.

Final Answer: 900 full-duplex voice channels

Interpretation

Using the values provided, the calculation gives **900 full-duplex voice channels**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 2

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.

SOLUTION

Formula: Channels per cell = Total channels / Number of cells in the reuse cluster.

Substitution: 560 / 7 = 80.

Final Answer: 80 channels per cell

Interpretation

Using the values provided, the calculation gives **80 channels per cell**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 3

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

SOLUTION

Formula: Simultaneous voice channels = Number of carriers x time slots per carrier.

Substitution: 100 x 8 = 800.

Final Answer: 800 simultaneous voice channels

Interpretation

Using the values provided, the calculation gives **800 simultaneous voice channels**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 4

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio (10·log10 of the ratio).

SOLUTION

Formula: Power ratio (dB) = 10 log10(P1 / P2).

Substitution: 10 log10(80 / 4) = 10 log10(20) = 13.01 dB.

Final Answer: Approximately 13.01 dB

Interpretation

Using the values provided, the calculation gives **Approximately 13.01 dB**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 5

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

SOLUTION

Formula: C/I (dB) = Carrier power (dBm) - Interference power (dBm).

Substitution: -60 - (-92) = -60 + 92 = 32 dB.

Final Answer: 32 dB

Interpretation

Using the values provided, the calculation gives **32 dB**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 6

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

SOLUTION

Formula: Total channels = Total bandwidth / Bandwidth per channel; channels per cell = Total channels / Reuse cluster size.

Substitution: 42,000 kHz / 300 kHz = 140 channels. 140 / 7 = 20.

Final Answer: 20 channels per cell

Interpretation

Using the values provided, the calculation gives **20 channels per cell**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 7

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

SOLUTION

Formula: P2 = P1 x (d1 / d2)^n, where n is the path-loss exponent.

Substitution: P2 = 64 x (1 / 2)^4 = 64 / 16 = 4 W.

Final Answer: 4 W

Interpretation

Using the values provided, the calculation gives **4 W**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 8

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

SOLUTION

For an inverse-square law, received power is proportional to transmit power divided by distance squared. Doubling distance reduces received power by a factor of 4, so transmit power must be multiplied by 2^2 = 4 to maintain the same edge SNR.

Final Answer: Increase transmit power by a factor of 4

Interpretation

Using the values provided, the calculation gives **Increase transmit power by a factor of 4**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 9

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

SOLUTION

Formula: Total channels in cluster = Cells per cluster x Channels per cell.

Substitution: 4 x 15 = 60.

Final Answer: 60 channels across the cluster

Interpretation

Using the values provided, the calculation gives **60 channels across the cluster**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 10

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

SOLUTION

Formula for maximum Doppler shift: fd = (v / c) x f, where c is approximately 3 x 10^8 m/s.

Substitution: v = 100 / 3.6 = 27.78 m/s. fd = (27.78 / 300,000,000) x 900,000,000 = 83.33 Hz.

Final Answer: Approximately 83.3 Hz maximum Doppler shift

Interpretation

Using the values provided, the calculation gives **Approximately 83.3 Hz maximum Doppler shift**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 11

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

SOLUTION

Formula: Slot duration = Frame duration / Number of slots.
Substitution: 5 ms / 8 = 0.625 ms.

Final Answer: 0.625 ms per slot

Interpretation

Using the values provided, the calculation gives **0.625 ms per slot**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 12

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

SOLUTION

Formula: Number of carriers = Total bandwidth / Carrier spacing.

Substitution: 24 MHz = 24,000 kHz. 24,000 / 200 = 120.

Final Answer: 120 carrier frequencies

Interpretation

Using the values provided, the calculation gives **120 carrier frequencies**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 13

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

SOLUTION

Convert transmit power to dBm: P(dBm) = 10 log10(P in mW). Thus, 8 W = 8,000 mW and Ptx = 10 log10(8,000) = 39.03 dBm.

Received power = Transmit power (dBm) - Path loss = 39.03 - 33 = 6.03 dBm.

Final Answer: Approximately 6.03 dBm

Interpretation

Using the values provided, the calculation gives **Approximately 6.03 dBm**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 14

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

SOLUTION

Formula: FSPL (dB) = 32.44 + 20 log10(f in MHz) + 20 log10(d in km).

Substitution: 32.44 + 20 log10(1900) + 20 log10(2.5) = 32.44 + 65.58 + 7.96 = 105.98 dB.

Final Answer: Approximately 106 dB free-space path loss over 2.5 km

Interpretation

Using the values provided, the calculation gives **Approximately 106 dB free-space path loss over 2.5 km**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 15

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

SOLUTION

Formula: Cluster capacity = Number of cells x Capacity per cell.

Substitution: 6 x 250 = 1,500.

Final Answer: 1,500 simultaneous calls

Interpretation

Using the values provided, the calculation gives **1,500 simultaneous calls**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 16

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

SOLUTION

Using the average arrival rate and call duration: Active calls = Call arrivals per second x Average duration in seconds.

Substitution: 4,000 x (3 x 60) = 4,000 x 180 = 720,000.

Final Answer: Approximately 720,000 active calls, assuming the stated arrival rate is sustained

Interpretation

Using the values provided, the calculation gives **Approximately 720,000 active calls, assuming the stated arrival rate is sustained**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 17

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

SOLUTION

Formula: Total bandwidth = Number of channels x Bandwidth per channel.

Substitution: 150 x 25 kHz = 3,750 kHz = 3.75 MHz.

Final Answer: 3.75 MHz

Interpretation

Using the values provided, the calculation gives **3.75 MHz**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 18

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

SOLUTION

If there are 30 frequency bands arranged in one consecutive row, there are 29 internal gaps between them. Total guard bandwidth = 29 x 40 kHz = 1,160 kHz = 1.16 MHz.

Final Answer: 1.16 MHz for 29 gaps

Note: If the question intends 30 separate guard bands, the result would be 30 x 40 kHz = 1.20 MHz.

Interpretation

Using the values provided, the calculation gives **1.16 MHz for 29 gaps**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 19

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

SOLUTION

Cell diameter = 2 x radius = 2 x 1.5 = 3 km. Distance travelled in one hour = 90 km.

Approximate handoffs = 90 km / 3 km per cell crossing = 30.

Final Answer: Approximately 30 handoffs per hour

Interpretation

Using the values provided, the calculation gives **Approximately 30 handoffs per hour**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 20

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

SOLUTION

Formula: Total capacity = Regional centers x Base stations per center x Users per base station.

Substitution: 8 x 40 x 250 = 80,000.

Final Answer: 80,000 users

Interpretation

Using the values provided, the calculation gives **80,000 users**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 21

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

SOLUTION

Convert radius: 400 m = 0.4 km. Using the approximation supplied in the question, cell area = R^2 = 0.4^2 = 0.16 km^2.

Number of cells = 400 / 0.16 = 2,500.

Final Answer: Approximately 2,500 cells

Interpretation

Using the values provided, the calculation gives **Approximately 2,500 cells**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 22

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

SOLUTION

Formula: Number of channels = Total spectrum / Bandwidth per channel.

Substitution: 6 MHz = 6,000 kHz. 6,000 / 30 = 200.

Final Answer: 200 channels

Interpretation

Using the values provided, the calculation gives **200 channels**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 23

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

SOLUTION

Formula: Offered traffic (Erlangs) = Subscribers x Calls per hour per subscriber x Average call duration in hours.

Substitution: 1,200 x 4 x (2.5 / 60) = 1,200 x 0.1667 = 200 Erlangs.

Final Answer: 200 Erlangs

Interpretation

Using the values provided, the calculation gives **200 Erlangs**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 24

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

SOLUTION

Formula: Maximum simultaneous users = Total bandwidth / Bandwidth per user.

Substitution: 40 MHz = 40,000 kHz. 40,000 / 400 = 100.

Final Answer: 100 simultaneous users

Interpretation

Using the values provided, the calculation gives **100 simultaneous users**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 25

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

SOLUTION

Formula: Total capacity = Number of base stations x Users per base station.

Substitution: 1,200 x 600 = 720,000.

Final Answer: 720,000 simultaneous users

Interpretation

Using the values provided, the calculation gives **720,000 simultaneous users**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 26

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

SOLUTION

Formula: Paging rounds = Total cells / Cells covered per round.

Substitution: 200 / 4 = 50.

Final Answer: 50 paging rounds

Interpretation

Using the values provided, the calculation gives **50 paging rounds**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 27

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

SOLUTION

Offered concurrent load = Call attempts per minute x Average call duration in minutes = 40,000 x 3 = 120,000 calls.

Excess over capacity = 120,000 - 100,000 = 20,000 calls. Blocked fraction = 20,000 / 120,000 x 100 = 16.67%.

Final Answer: Approximately 16.67% of the simplified offered load is blocked

This uses the direct capacity-overflow approximation implied by the question.

Interpretation

Using the values provided, the calculation gives **Approximately 16.67% of the simplified offered load is blocked**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 28

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A network policy lets 15% of its 600,000 subscribers roam onto partner networks each hour. At any given moment, how many subscribers are likely roaming?

SOLUTION

Formula: Roaming subscribers = Total subscribers x Roaming percentage.

Substitution: 600,000 x (15 / 100) = 600,000 x 0.15 = 90,000.

Final Answer: 90,000 subscribers

Interpretation

Using the values provided, the calculation gives **90,000 subscribers**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 29

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A switching center is capable of processing 12,000 calls per second, but is currently handling a peak load of 9,000 calls per second. What percentage of its capacity is being utilized?

SOLUTION

Formula: Utilization (%) = Current load / Total capacity x 100.

Substitution: 9,000 / 12,000 x 100 = 75%.

Final Answer: 75% utilization

Interpretation

Using the values provided, the calculation gives **75% utilization**. This is the estimated result for the mobile network scenario described in the question.

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ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 30

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

An operator reuses frequencies with a factor of 4 across a network holding 48 MHz of spectrum, with 200 kHz per channel. How many channels land in each cell?

SOLUTION

Total channels = 48 MHz / 200 kHz = 48,000 / 200 = 240 channels.

Channels per cell = 240 / Reuse factor 4 = 60.

Final Answer: 60 channels per cell

Interpretation

Using the values provided, the calculation gives **60 channels per cell**. This is the estimated result for the mobile network scenario described in the question.

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Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 31

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A 900 MHz GSM deployment uses 200 kHz channel spacing across a 30 MHz allocation. How many channels can this spectrum support?

SOLUTION

Formula: Number of channels = Total bandwidth / Channel spacing.

Substitution: 30 MHz = 30,000 kHz. 30,000 / 200 = 150.

Final Answer: 150 channels

Interpretation

Using the values provided, the calculation gives **150 channels**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 32

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

Over the course of an hour, a network logs 2500 handoff attempts, 99% of which succeed. How many calls were successfully handed off?

SOLUTION

Formula: Successful handoffs = Total attempts x Success rate.

Substitution: 2,500 x 0.99 = 2,475.

Final Answer: 2,475 successful handoffs

Interpretation

Using the values provided, the calculation gives **2,475 successful handoffs**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 33

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A mobile handset radiates 12 W, and the link experiences 45 dB of path loss before reaching the base station. What power level does the base station receiver detect?

SOLUTION

Convert transmit power to dBm: 12 W = 12,000 mW, so Ptx = 10 log10(12,000) = 40.79 dBm.
Received power = 40.79 - 45 = -4.21 dBm.

Final Answer: Approximately -4.21 dBm

Interpretation

Using the values provided, the calculation gives **Approximately -4.21 dBm**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 34

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A 7-cell cluster is deployed where each cell can serve up to 900 simultaneous users. What is the cluster's overall user capacity?

SOLUTION

Formula: Cluster capacity = Number of cells x Users per cell.

Substitution: 7 x 900 = 6,300.

Final Answer: 6,300 simultaneous users

Interpretation

Using the values provided, the calculation gives **6,300 simultaneous users**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 35

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A TDMA carrier's frame spans 4.8 ms and is split into 8 equal slots. How long is each individual slot?

SOLUTION

Formula: Slot duration = Frame duration / Number of slots.
Substitution: 4.8 ms / 8 = 0.6 ms.

Final Answer: 0.6 ms per slot

Interpretation

Using the values provided, the calculation gives **0.6 ms per slot**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 36

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A mobile device moves at 150 km/h while connected to a 1.8 GHz carrier. What Doppler shift should the receiver expect?

SOLUTION

Formula for maximum Doppler shift: fd = (v / c) x f.

Substitution: v = 150 / 3.6 = 41.67 m/s. fd = (41.67 / 300,000,000) x 1,800,000,000 = 250 Hz.

Final Answer: Approximately 250 Hz maximum Doppler shift

Interpretation

Using the values provided, the calculation gives **Approximately 250 Hz maximum Doppler shift**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 37

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A core switch is dimensioned for 250,000 simultaneous calls. It is fielding calls at a rate of 70,000 per minute, each averaging 4 minutes in duration. What share of the offered load gets blocked?

SOLUTION

Offered concurrent load = Call attempts per minute x Average duration = 70,000 x 4 = 280,000 calls.

Excess over capacity = 280,000 - 250,000 = 30,000 calls. Blocked share = 30,000 / 280,000 x 100 = 10.71%.

Final Answer: Approximately 10.71% of the simplified offered load is blocked

Interpretation

Using the values provided, the calculation gives **Approximately 10.71% of the simplified offered load is blocked**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 38

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A hexagonal layout groups 9 cells into a cluster, and each cell can host 600 users. What is the cluster's combined user capacity?

SOLUTION

Formula: Cluster capacity = Cells per cluster x Users per cell.

Substitution: 9 x 600 = 5,400.

Final Answer: 5,400 users

Interpretation

Using the values provided, the calculation gives **5,400 users**. This is the estimated result for the mobile network scenario described in the question.

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 39

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A switching center processes 60,000 calls per second, with 0.4% of them dropped due to network issues. How many calls are being dropped every second?

SOLUTION

Formula: Dropped calls per second = Total calls per second x Drop percentage.

Substitution: 60,000 x (0.4 / 100) = 60,000 x 0.004 = 240.

Final Answer: 240 calls dropped per second

Interpretation

Using the values provided, the calculation gives **240 calls dropped per second**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 40

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A base station in an 1800 MHz network transmits at 43 dBm. At the edge of its 2 km-radius cell, the received signal measures -87 dBm. What path loss has the signal undergone?

SOLUTION

Formula: Path loss (dB) = Transmit power (dBm) - Received power (dBm).

Substitution: 43 - (-87) = 43 + 87 = 130 dB.

Final Answer: 130 dB path loss

Interpretation

Using the values provided, the calculation gives **130 dB path loss**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 41

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A single base station covers 2.5 km² of territory, and the operator plans to deploy 150 identical stations. What total geographic area will the rollout cover?

SOLUTION

Formula: Total coverage area = Area per station x Number of stations.

Substitution: 2.5 km^2 x 150 = 375 km^2.

Final Answer: 375 km^2

Interpretation

Using the values provided, the calculation gives **375 km^2**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 42

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

An operator applies a frequency reuse factor of 6 over a 48 MHz spectrum allocation, with each channel taking up 200 kHz. How many channels does each cell get?

SOLUTION

Total channels = 48 MHz / 200 kHz = 48,000 / 200 = 240 channels.

Channels per cell = 240 / 6 = 40.

Final Answer: 40 channels per cell

Interpretation

Using the values provided, the calculation gives **40 channels per cell**. This is the estimated result for the mobile network scenario described in the question.

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 - SHORT ANSWERS

Answer to Question 43

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A paging system processes 6000 users per cycle, and the subscriber base totals 150,000 users. How many paging cycles are needed to sweep the entire base?

SOLUTION

Formula: Paging cycles = Total subscribers / Users processed per cycle.

Substitution: 150,000 / 6,000 = 25.

Final Answer: 25 paging cycles

Interpretation

Using the values provided, the calculation gives **25 paging cycles**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 44

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A phone call runs for 9 minutes, and during that time the network hands the call off roughly every 3 minutes. How many handoffs took place over the call?

SOLUTION

Using the interval-based estimate: Handoffs = Call duration / Handoff interval = 9 / 3 = 3.

Final Answer: Approximately 3 handoffs using the intended interval calculation

Timing note: If the call ends exactly at minute 9 and a handoff at the instant of termination is not counted, only the handoffs at minutes 3 and 6 would have completed before the call ended.

Interpretation

Using the values provided, the calculation gives **Approximately 3 handoffs using the intended interval calculation**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 45

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A receiver picks up a wanted signal at -85 dBm against a noise floor of -115 dBm. What signal-to-noise ratio, in dB, does this represent?

SOLUTION

Formula: SNR (dB) = Signal power (dBm) - Noise power (dBm).
Substitution: -85 - (-115) = -85 + 115 = 30 dB.

Final Answer: 30 dB SNR

Interpretation

Using the values provided, the calculation gives **30 dB SNR**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 46

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A 4G cell is serving 250 users simultaneously, each needing a guaranteed 150 kbps. What is the minimum bandwidth the cell must provide to support them all?

SOLUTION

Aggregate required data rate = Number of users x Data rate per user.

Substitution: 250 x 150 kbps = 37,500 kbps = 37.5 Mbps.

Final Answer: At least 37.5 Mbps of aggregate throughput

Technical note: Converting this data rate to radio-frequency bandwidth in MHz also requires the system's spectral efficiency. The question's supplied values directly determine the required aggregate data rate.

Interpretation

Using the values provided, the calculation gives **At least 37.5 Mbps of aggregate throughput**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 47

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A base station handles 1500 subscribers, each averaging 4 calls per hour with a 2.5-minute call length. What total traffic, in Erlangs, does this generate?

SOLUTION

Formula: Traffic (Erlangs) = Subscribers x Calls per hour per subscriber x Average call duration in hours.
Substitution: 1,500 x 4 x (2.5 / 60) = 1,500 x 0.1667 = 250 Erlangs.

Final Answer: 250 Erlangs

Interpretation

Using the values provided, the calculation gives **250 Erlangs**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 48

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A network reserves 250 separate control channels, each 12 kHz wide, for signaling. What total bandwidth is tied up in control channels?

SOLUTION

Formula: Total control bandwidth = Number of channels x Bandwidth per channel.

Substitution: 250 x 12 kHz = 3,000 kHz = 3 MHz.

Final Answer: 3 MHz

Interpretation

Using the values provided, the calculation gives **3 MHz**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 49

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A network is carrying 1500 simultaneous connections, each streaming at 600 kbps on average. What total throughput must the network support?

SOLUTION

Formula: Total throughput = Number of simultaneous connections x Throughput per connection.

Substitution: 1,500 x 600 kbps = 900,000 kbps = 900 Mbps = 0.9 Gbps.

Final Answer: 900 Mbps (0.9 Gbps)

Interpretation

Using the values provided, the calculation gives **900 Mbps (0.9 Gbps)**. This is the estimated result for the mobile network scenario described in the question.

Answer to Question 50

Student Name: NAVEEN.S Reg. No.: 192324248

QUESTION

A telecom operator wants each tower to cover a 4 km radius (treating coverage as a circle, area = pi x R^2) and needs to blanket a 350 km^2 service region. Roughly how many towers will the rollout require?

SOLUTION

Area per tower = pi x R^2 = pi x 4^2 = 16 pi, approximately 50.27 km^2.

Number of towers = 350 / 50.27 = approximately 6.96. Round up to a whole tower: 7.

Final Answer: Approximately 7 towers, using the ideal circular-area estimate

Interpretation

Using the values provided, the calculation gives **Approximately 7 towers, using the ideal circular-area estimate**. This is the estimated result for the mobile network scenario described in the question.